Problem D.5

Find: • For the eom's
(a) s-domain solution
(b) transfer function $H(s) = \frac{Z(s)}{F(s)}$
and associated block diagram (c)

Solution:

Using sympy for $H = C(sI - A)^{-1}B + D$...

$$Z(s) = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}} F(s) \quad \text{(a)}$$

which implies...

$$H(s) = \frac{Z(s)}{F(s)} = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}} \quad \text{(b)}$$

So...

$$F(s) \longrightarrow \boxed{H(s)} \xrightarrow{Z(s)} \quad \text{(c)}$$

Problem E.5

Find: • For linearized eom's...
(a) s-domain solutions
(b) $H_1(s) = \frac{\tilde{Z}(s)}{\tilde{F}(s)}$, $H_2(s) = \frac{\tilde{\Theta}(s)}{\tilde{F}(s)}$, $H_3(s) = \frac{\tilde{Z}(s)}{\tilde{\Theta}(s)}$
(c) Assume $m_1 g \tilde{z}$ is negligible in $\ddot{\tilde{\theta}}(t)$. How does this simplify $H_2(s)$ and $H_3(s)$?
(d) Block diagram from $\tilde{F}(s)$ to $\tilde{\Theta}(s)$ to $\tilde{Z}(s)$.

Solution:

Using sympy...

$$\tilde{Z}(s) = \frac{-3lg}{(m_2 l^2 + 3m_1 z_e^2)s^4 - 3m_1 g^2} \tilde{F}(s) \quad (\tilde{Z}(s))$$

$$\tilde{\Theta}(s) = \frac{3ls^2}{(m_2 l^2 + 3m_1 z_e^2)s^4 - 3m_1 g^2} \tilde{F}(s) \quad (\tilde{\Theta}(s))$$

(a)

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Then we have ...

$$H_1(s) = \frac{\tilde{Z}(s)}{\tilde{F}(s)} = \frac{-3lg}{(m_2 l^2 + 3m_1 z_e^2)s^4 - 3m_1 g^2} \leftarrow H_1(s)$$

$$H_2(s) = \frac{\tilde{\Theta}(s)}{\tilde{F}(s)} = \frac{3ls^2}{(m_2 l^2 + 3m_1 z_e^2)s^4 - 3m_1 g^2} \leftarrow H_2(s)$$

(b)

$$H_3(s) = \frac{\tilde{Z}(s)}{\tilde{\Theta}(s)} = \frac{\tilde{Z}(s)}{\tilde{F}(s)} \cdot \frac{\tilde{F}(s)}{\tilde{\Theta}(s)} = H_1(s) \cdot \frac{1}{H_2(s)} = \frac{-3lg}{3ls^2} = -\frac{g}{s^2} \leftarrow H_3(s)$$

Then if the $m_1 g \tilde{z}$ term goes to zero in our formula for $\ddot{\tilde{z}}(t)$, then we can set the entry $A[3,0] = 0$, which yields...

$$H_1(s) = \frac{-3lg}{m_2 l^2 s^4 + 3m_1 z_e^2 s^4}$$

$$H_2(s) = \frac{3l}{m_2 l^2 s^2 + 3m_1 z_e^2 s^2}$$

$$H_2(s) = \frac{H_1(s)}{H_2(s)} = \frac{-3lg}{3l} = -g$$

(c)

Then the block diagram is...

$$\tilde{F}(s) \rightarrow \boxed{H_2(s)} \xrightarrow{\tilde{\theta}(s)} \boxed{H_3(s)} \rightarrow \tilde{Z}(s)$$

(d)

Problem F.5

Find: For eom's...

(a) s-domain solutions

(b) For longitudinal, find transfer function from input $F(s)$ to $H(s)$.

(c) For lateral, find from input $\tilde{T}(s)$ to intermediate $\tilde{\theta}(s)$ and output $\tilde{z}(s)$. Find T.F. from $\tilde{\theta}(s)$ to $\tilde{z}(s)$.

(d) Draw block diagrams for longitudinal & lateral systems.

Solution:

Using sympy, where $F_e = g(m_c + m_l + m_r)$

$$\tilde{Z}(s) = \frac{F_e}{(J_c(m_c + 2m_r) + 2m_c m_r d^2 + 4m_r^2 d^2)s^4 + (J_c \mu + 2m_r \mu d^2)s^3} \tilde{T}(s)$$

(a)

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$$\tilde{H}(s) = \frac{1}{(m_c + 2m_r)s^2}\tilde{F}(s)$$

$$\tilde{\Theta}(s) = \frac{1}{(J_c + 2m_r d^2)s^2}\tilde{T}(s)$$

(a)

Then for longitudinal...

$$H_1(s) = \left(\frac{\tilde{H}(s)}{\tilde{F}(s)}\right) = \frac{1}{(m_c + 2m_r)s^2}$$

(b)

For lateral...

$$H_2(s) = \frac{\tilde{\Theta}(s)}{\tilde{T}(s)} = \frac{1}{(J_c + 2m_r d^2)s^2}$$

$$H_3(s) = \frac{\tilde{Z}(s)}{\tilde{T}(s)} = \frac{F_e}{((m_c + 2m_r)J_c + (2m_c m_r + 4m_r^2)d^2)s^4 + \mu(J_c + 2m_r d^2)s^3}$$

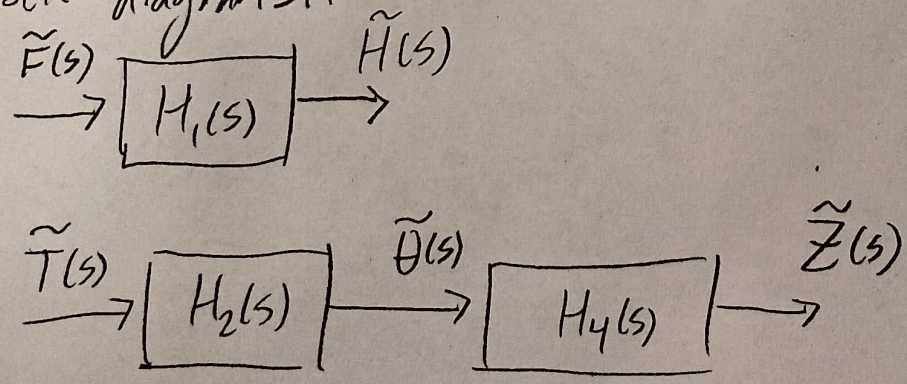
(c)

$$H_4(s) = \frac{\tilde{Z}(s)}{\tilde{\Theta}(s)} = \frac{\tilde{Z}(s)}{\tilde{T}(s)}\frac{\tilde{T}(s)}{\tilde{\Theta}(s)} = H_3(s)\cdot\frac{1}{H_2(s)}$$

$$= \frac{F_e(J_c + 2m_r d^2)s^2}{((m_c + 2m_r)J_c + (2m_c m_r + 4m_r^2)d^2)s^4 + \mu(J_c + 2m_r d^2)s^3}$$

$$= \frac{F_e(J_c + 2m_r d^2)}{((m_c + 2m_r)J_c + (2m_c m_r + 4m_r^2)d^2)s^2 + \mu(J_c + 2m_r d^2)s}$$

Block diagrams...



(d)